Test 3 Practice — Circuits

These are modeled on the hardest problems from Written HW 5 and HW 6 – same problem types, new numbers. Each pair of problems below tests the same idea but asks for a different unknown, so skim the whole question before you start plugging in. The answer key on the last page.

Part A — Series & Parallel Resistors

1. Three resistors, $R_1 = 1.0\ \text{k}\Omega$, $R_2 = 1.5\ \text{k}\Omega$, and $R_3 = 2.5\ \text{k}\Omega$, are in series with a battery of $\varepsilon = 9.0$ V. Find (a) the equivalent resistance, and (b) the current supplied by the battery.

2. Three resistors are in series with a battery of $\varepsilon = 12$ V. Two of them are known: $R_1 = 2.0\ \text{k}\Omega$ and $R_2 = 3.0\ \text{k}\Omega$. The current supplied by the battery is measured to be 1.5 mA. Find the third resistor R_3.

3. Two resistors are in parallel across a battery of $\varepsilon = 15$ V. $R_1 = 3.0$ kΩ, and the *total* current measured from the battery is 8.0 mA. Find (a) the current through R_2, and (b) the value of R_2.

4. Two resistors, $R_1 = 2.0$ kΩ and $R_2 = 6.0$ kΩ, are in parallel across a battery of $\varepsilon = 18$ V. Find (a) the equivalent resistance, and (b) the total current supplied by the battery.

Part B — Real Batteries, RC Circuits, & Power

5. A real battery is connected to a variable resistor R. When $R = 0$ (a dead short), the current is 6.0 A. When $R = 2.0\ \Omega$, the current drops to 4.0 A. Find the battery's emf ε and its internal resistance r.

6. A battery has emf $\varepsilon = 20$ V and internal resistance $r = 2.0\ \Omega$. What external resistance R should be connected to the battery so that the current is 2.5 A?

7. You have two resistors, $R_1 = 1.0\ \text{k}\Omega$ and $R_2 = 4.0\ \text{k}\Omega$, and two capacitors, $C_1 = 2.0\ \mu\text{F}$ and $C_2 = 3.0\ \mu\text{F}$. (a) If all four are wired onto the same two rails (parallel), find the time constant for discharging the combination. (b) If instead all four are wired into a single series loop, find the time constant.

8. Two resistors, $R_1 = 1.0\ \text{k}\Omega$ and $R_2 = 5.0\ \text{k}\Omega$, and a capacitor $C_1 = 4.0\ \mu\text{F}$ are wired into a single series loop along with a second capacitor C_2. The measured time constant for this series combination is $\tau = 18$ ms. Find C_2.

9. The switch in an RC circuit has been in the "charge" position for a long time, then flips to the "discharge" position at $t = 0$, disconnecting the battery and leaving the capacitor to discharge through the resistor alone. The battery was $\varepsilon = 8.0$ V, $C = 5.0$ μF, and $R = 40$ Ω. Find the charge Q on the capacitor and the current I through the resistor (a) immediately after the switch flips, (b) at $t = 100$ μs, and (c) at $t = 400$ μs.

10. The switch in an RC circuit flips from "charge" to "discharge" at $t = 0$. The battery was $\varepsilon = 6.0$ V and $C = 3.0$ μF, and the measured time constant for the discharge is $\tau = 150$ μs. Find (a) the resistance R, and (b) the charge Q and current I at $t = 300$ μs.

11. You have three resistors — 2.0 kΩ, 4.0 kΩ, and 8.0 kΩ — and a 24 V supply. Find the ratio of the total power delivered to the resistors when they're connected in parallel to the total power delivered when they're connected in series.

12. Three resistors — 1.0 kΩ, 3.0 kΩ, and 6.0 kΩ — are connected in series across a power supply. The supply delivers a total power of 40 mW to this series combination. Find (a) the supply voltage ΔV, and (b) the total power delivered if the same three resistors were instead connected in parallel across that same voltage.

ANSWER KEY — FLIP THE PAGE OVER (OR STAND ON YOUR HEAD)

Part A — Series & Parallel Resistors

1. $R_{\text{eq}} = 5.0\ \text{k}\Omega$; $I = 1.8\ \text{mA}$
2. $R_3 = 3.0\ \text{k}\Omega$
3. $I_2 = 3.0\ \text{mA}$; $R_2 = 5.0\ \text{k}\Omega$
4. $R_{\text{eq}} = 1.5\ \text{k}\Omega$; $I_{\text{tot}} = 12\ \text{mA}$

Part B — Real Batteries, RC Circuits, & Power

5. $r = 4.0\ \Omega$; $\varepsilon = 24\ \text{V}$
6. $R = 6.0\ \Omega$
7. (a) parallel: $\tau = 4.0\ \text{ms}$ (b) series: $\tau = 6.0\ \text{ms}$
8. $C_2 = 12.0\ \mu\text{F}$
9. (a) $Q_0 = 40\ \mu\text{C}$, $I_0 = 200\ \text{mA}$ (b) $t = 100\ \mu\text{s}$: $Q = 24.3\ \mu\text{C}$, $I = 121\ \text{mA}$ (c) $t = 400\ \mu\text{s}$: $Q = 5.41\ \mu\text{C}$, $I = 27.1\ \text{mA}$
10. (a) $R = 50\ \Omega$ (b) $Q = 2.44\ \mu\text{C}$, $I = 16.2\ \text{mA}$
11. $P_{\text{parallel}} \approx 504\ \text{mW}$; $P_{\text{series}} \approx 41.1\ \text{mW}$; ratio ≈ 12.25
12. (a) $\Delta V = 20\ \text{V}$ (b) $P_{\text{parallel}} = 600\ \text{mW}$